

LAB SESSION 07

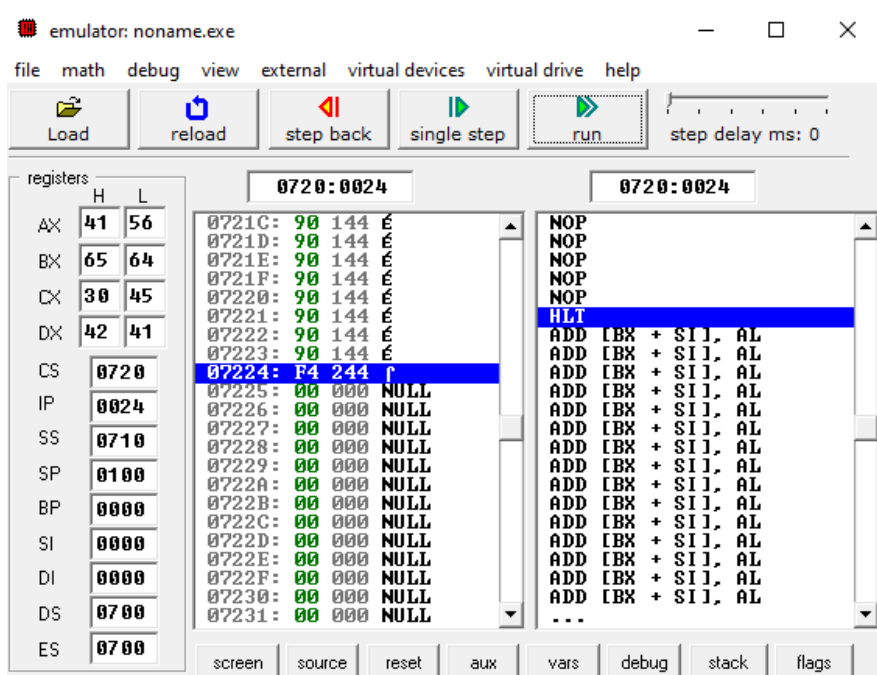
BASIC ARITHMETIC OPERATIONS

1. Program to add two 32-bit numbers 23231111h and 42413045h, stored in BX:AX and DX:CX, respectively is shown below. The given program uses ADD and ADC (add with carry) instructions. Run the program and complete the following Table. Also analyze the flag affected.

Source Code:

```
01 TITLE LAB 7 ACTIVITY 1           ;Title of the program
02 .MODEL SMALL                     ;Dedine memory model
03 .STACK 100H                      ;Set the size of the stack
04 .CODE                            ;Code segment starts
05
06 MAIN PROC                        ;Start of the main procedure
07     MOV AX,1111H                  ;Move the hexadecimal value 1111H into the AX register
08     MOV BX,2323H                  ;Move the hexadecimal value 2323H into the BX register
09     MOV CX,3045H                  ;Move the hexadecimal value 3045H into the CX register
10     MOV DX,4241H                  ;Move the hexadecimal value 4241H into the DX register
11
12     ADD AX,CX                     ;Add the contents of CX to AX
13     ADC BX,DX                     ;Add the contents of DX and the carry flag to BX
14
15 MAIN ENDP                        ;End of the main procedure
16 END MAIN                          ;End of the program with the label MAIN
```

OUTPUT:



2. Program to subtract two 32-bit numbers 20003000h and 40001000h, stored in BX:AX and DX:CX, respectively is shown below. The given program uses SUB and SBB (subtract with borrow) instructions. Run the program and complete the following Table. Also analyze the flag affected.

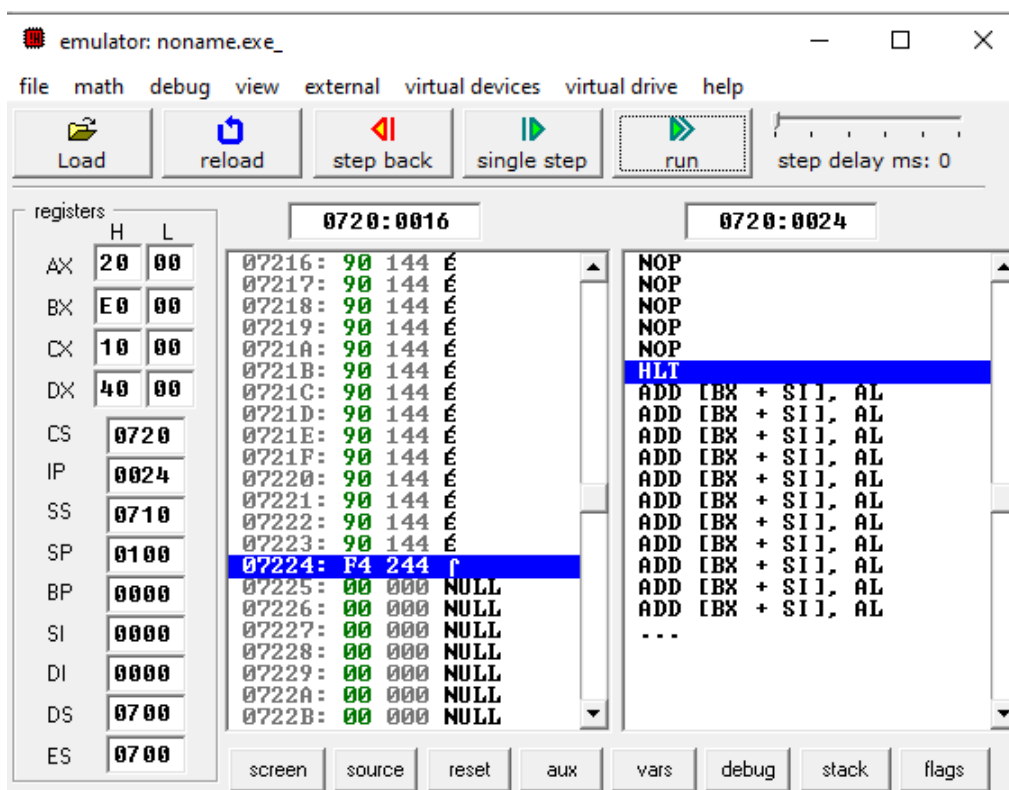
Source Code:

```

01 TITLE LAB 7 ACTIVITY 2          ;Title of the program
02 .MODEL SMALL                    ;Dedine memory model
03 .STACK 100H                     ;Set the size of the stack
04 .CODE                           ;Code segment starts
05
06 MAIN PROC                       ;Start of the main procedure
07     MOV AX,3000H                 ;Move the hexadecimal value 3000H into the AX register
08     MOV BX,2000H                 ;Move the hexadecimal value 2000H into the BX register
09     MOV CX,1000H                 ;Move the hexadecimal value 1000H into the CX register
10     MOV DX,4000H                 ;Move the hexadecimal value 4000H into the DX register
11
12     SUB AX,CX                    ;Subtract the contents of CX to AX
13     SBB BX,DX                    ;Subtract the contents of DX and the Borrow flag from BX
14
15 MAIN ENDP                       ;End of the main procedure
16 END MAIN                        ;End of the program with the label MAIN

```

OUTPUT:



3. Program to multiply two decimal numbers 5 and 10, stored register BL and CL respectively is shown below using the MUL instructions. The result is stored in a register DX. Run the program and complete the following Table. Also analyze the flag affected.

Source Code:

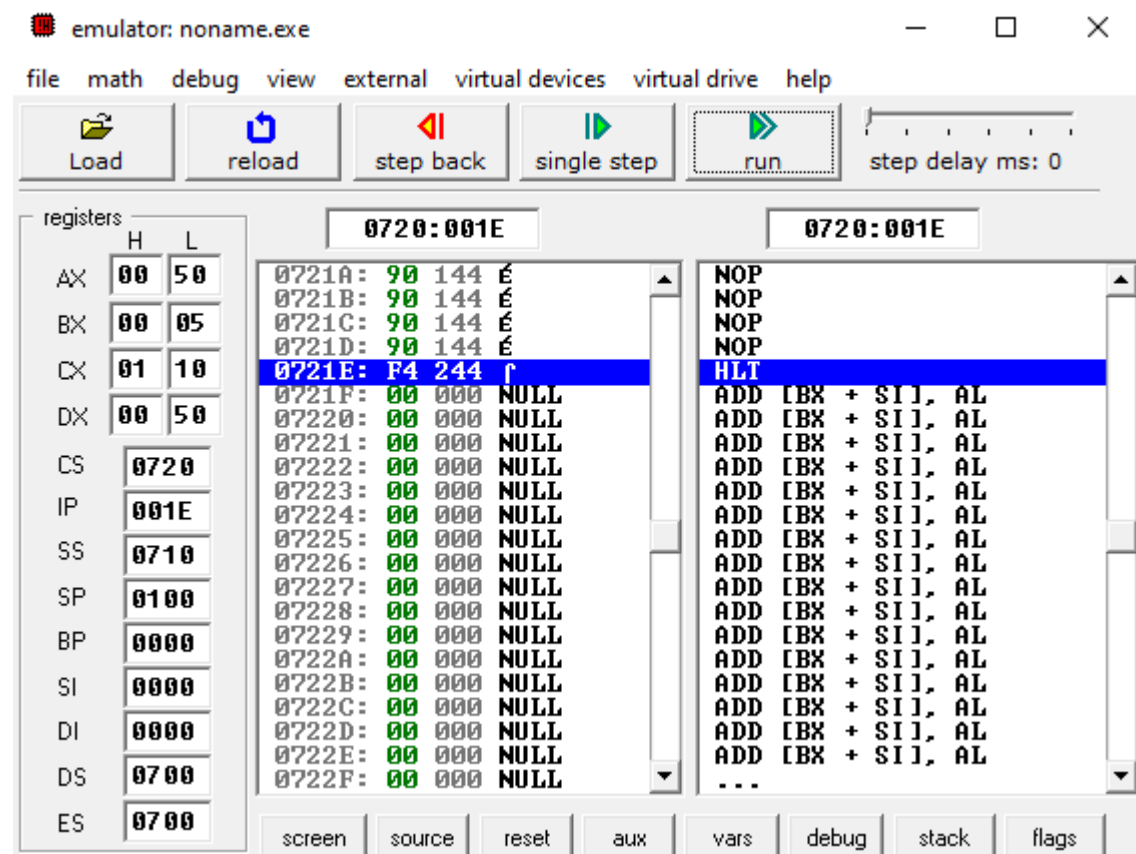
OUTPUT:

```

01 TITLE LAB 7 ACTIVITY 3                ;Title of the program
02 .MODEL SMALL                          ;Dedine memory model
03 .STACK 100H                           ;Set the size of the stack
04 .CODE                                 ;Code segment starts
05
06 MAIN PROC                             ;Start of the main procedure
07     MOV BL,05H                         ;Move the hexadecimal value 05H into the BL register
08     MOV CL,10H                         ;Move the hexadecimal value 10H into the CL register
09     MOV AL,CL                          ;Move the contents of CL into the AL register
10
11     MUL BL                             ;Multiply AL by the contents of BL, result in AX
12     MOV DX,AX                          ;Move the contents of AX into the DX register
13
14 MAIN ENDP                             ;End of the main procedure
15 END MAIN                              ;End of the program with the label MAIN

```

OUTPUT:



4. Program to divide a decimal numbers 10 by a decimal number 5, stored register CL and BL respectively is shown below using DIV instructions. The result is stored in a register DX. Run the program and complete the following Table. Also analyze the flag affected.

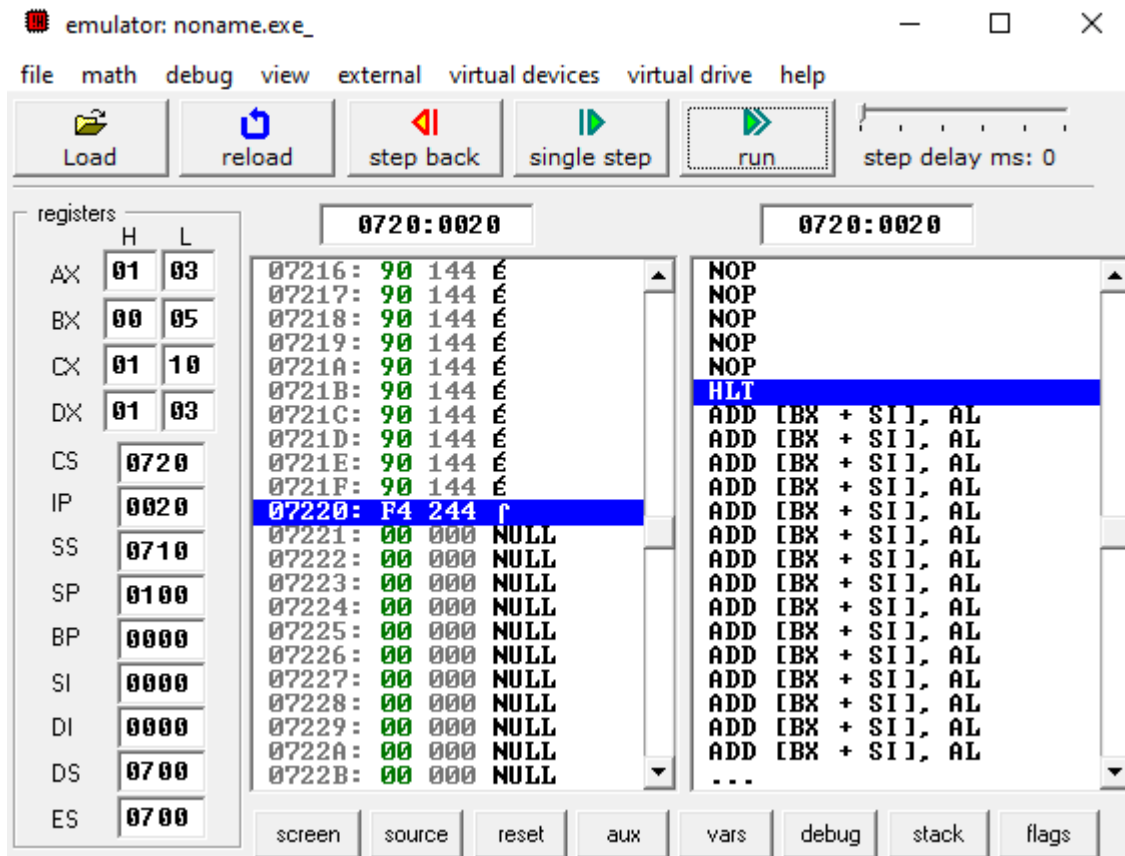
Source Code:

```

01 TITLE LAB 7 ACTIVITY 4                ;Title of the program
02 .MODEL SMALL                          ;Dedine memory model
03 .STACK 100H                           ;Set the size of the stack
04 .CODE                                 ;Code segment starts
05
06 MAIN PROC                             ;Start of the main procedure
07     MOV BL,05H                         ;Load BL register with the value 05H
08     MOV CL,10H                         ;Load CL register with the value 10H
09     MOV AH,00                          ;Clear the AH register
10     MOV AL,CL                           ;Load AL register with the contents of CL
11
12     DIV BL                             ;Divide the contents of the DX:AX register pair by BL
13     MOV DX,AX                           ;Move the contents of AX into the DX register
14
15 MAIN ENDP                             ;End of the main procedure
16 END MAIN                             ;End of the program with the label MAIN

```

OUTPUT:



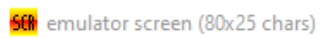
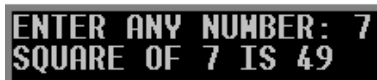
5. Write an Assembly language program to take input number from user and find the square of a number

Source Code:

```

01 TITLE LAB 7 ACTIVITY 5          ;Title of the program
02 .MODEL SMALL                   ;Define memory model
03 .STACK 100H                    ;Set the size of the stack
04 .DATA                           ;Data segment starts
05 MSG DB "ENTER ANY NUMBER: $"   ;Define a string message to prompt the user for input
06 MSG1 DB 0AH, 0DH, "SQUARE OF ", "$" ;Define a multi-line string message
07 MSG2 DB " IS $"                ;Define a string message indicating the end of the second line
08
09 .CODE                           ;Code segment starts
10 MAIN PROC                       ;Start of the main procedure
11     MOV AX, @DATA                ;Load the address of the data segment into AX
12     MOV DS, AX                   ;Set the data segment DS to the value in AX
13
14 AGAIN:
15     MOV AH, 9                    ;Set the interrupt function to display a string
16     LEA DX, MSG                  ;Load the offset of MSG into DX
17     INT 21H                      ;Call interrupt 21H to display the string
18
19     MOV AH, 1                    ;Set the interrupt function to read a character
20     INT 21H                      ;Call interrupt 21H to read a character into AL
21
22     PUSH AX                      ;Save the entered character on the stack
23
24     CMP AL, 39H                  ;Compare AL with the ASCII value for '9'
25     JG AGAIN                     ;Jump to AGAIN if AL is greater than '9'
26     CMP AL, 30H                  ;Compare AL with the ASCII value for '0'
27     JL AGAIN                     ;Jump to AGAIN if AL is less than '0'
28
29     PUSH AX                      ;Save the entered character on the stack
30     SUB AL, 30H                  ;Convert ASCII character to integer
31     MOV BL, AL                   ;Move the integer value to BL
32     MUL BL                       ;Multiply AX by BL
33     AAM                          ;ASCII adjust AX after multiplication
34     MOV BX, AX                   ;Move the result of the multiplication to BX
35
36     MOV AH, 9                    ;Set the interrupt function to display a string
37     LEA DX, MSG1                  ;Load the offset of MSG1 into DX
38     INT 21H                      ;Call interrupt 21H to display the string
39
40     POP DX                       ;Retrieve the entered character from the stack
41     MOV AH, 2                    ;Set the interrupt function to display a character
42     INT 21H                      ;Call interrupt 21H to display the character
43
44     MOV AH, 9                    ;Set the interrupt function to display a string
45     LEA DX, MSG2                  ;Load the offset of MSG2 into DX
46     INT 21H                      ;Call interrupt 21H to display the string
47
48     MOV DL, BH                   ;Move the high part of the result to DL
49     ADD DL, 30H                  ;Convert DL to ASCII
50     MOV AH, 02H                  ;Set the interrupt function to display a character
51     INT 21H                      ;Call interrupt 21H to display the character
52
53     MOV DL, BL                   ;Move the low part of the result to DL
54     ADD DL, 30H                  ;Convert DL to ASCII
55     MOV AH, 02H                  ;Set the interrupt function to display a character
56     INT 21H                      ;Call interrupt 21H to display the character
57
58     MOV AH, 4CH                  ;Set the interrupt function to exit the program
59     INT 21H                      ;Call interrupt 21H to exit the program
60
61 MAIN ENDP                       ;End of the main procedure
62 END MAIN                         ;End of the program with the label MAIN

```

OUTPUT: emulator screen (80x25 chars)

```
ENTER ANY NUMBER: 7  
SQUARE OF 7 IS 49
```